



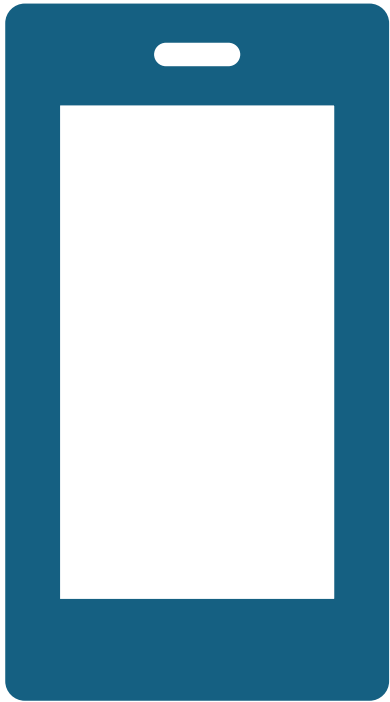
Tiffin

Order Alert System

Problem Statement

- Order notifications can be missed or cease after a short duration
- No alerts are received if the POS browser tab is closed
- Alerts are not persistent until an order is accepted
- No escalation mechanism exists for unattended orders
- Inefficient handling of orders across multiple branches

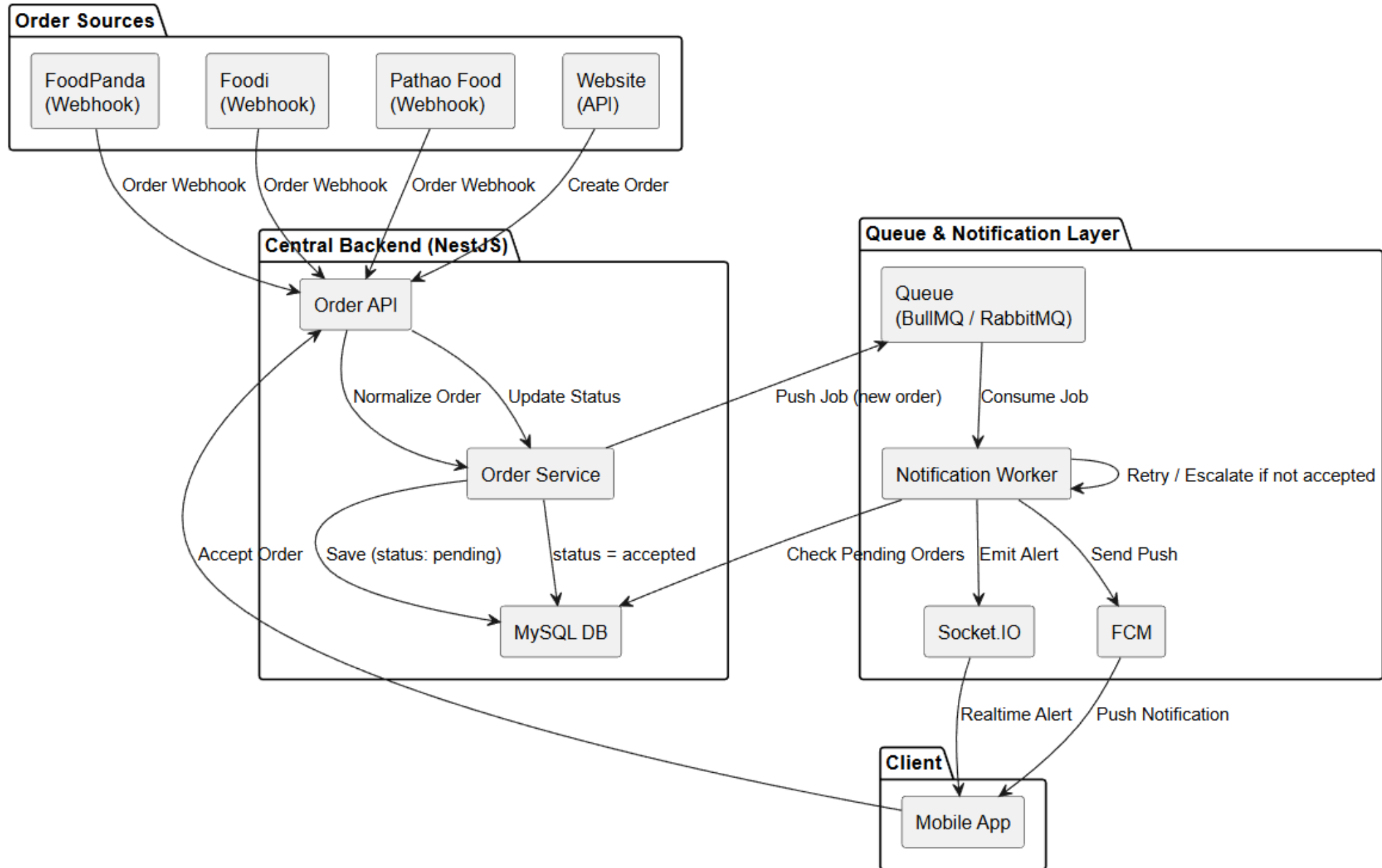




Assessment Scope

- Capture and normalize orders from multiple sources
- Deliver real-time alerts to branch devices (mobile and web)
- Ensure continuous alerting until order acceptance
- Provide a dashboard to monitor order status (pending / accepted)
- Maximum order routing delay: 10 seconds
- Support multi-branch operations with branch-specific routing

Order Alerting Flow (Simplified with Queue)



Development Stack

- **Backend:** NestJS
- **Database:** MySQL
- **Real-time Communication:** Socket.IO
- **Push Notifications:** Firebase Cloud Messaging (FCM)
- **Queue / Async Processing:** BullMQ (Redis) / RabbitMQ (notifications, retries, escalation)
- **Frontend Dashboard:** Next.js (Admin & monitoring panel)
- **Mobile App:** React Native
- **Infrastructure:** Dockerized deployment on VPS/Cloud

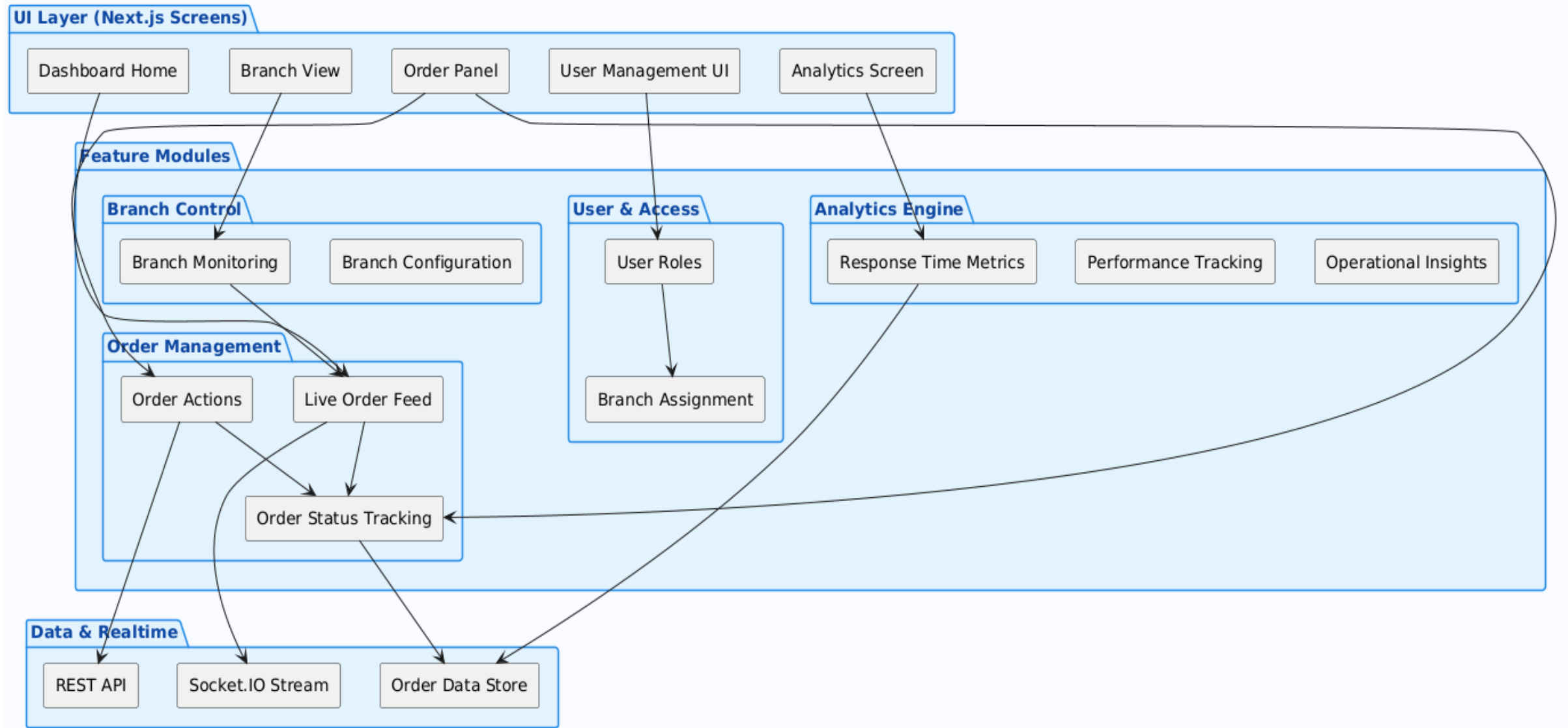




UI Dashboard Features

- **Live Order Feed**
Real-time incoming orders per branch via Socket.IO
- **Order Status Tracking**
End-to-end lifecycle visibility: Pending → Accepted → Escalated
- **Branch-wise Monitoring**
Separate dashboards for each branch with filtered views
- **Order Actions**
Accept / manage orders directly from dashboard interface
- **User Management**
Assign users to specific branches and manage app users (roles & access)
- **Branch Management**
Sync, configure, and maintain branch-level settings
- **Order Lifecycle Analytics**
 - Average response time per branch
 - User-wise performance tracking
 - Operational efficiency insights

Dashboard UI (Layered Feature Architecture)

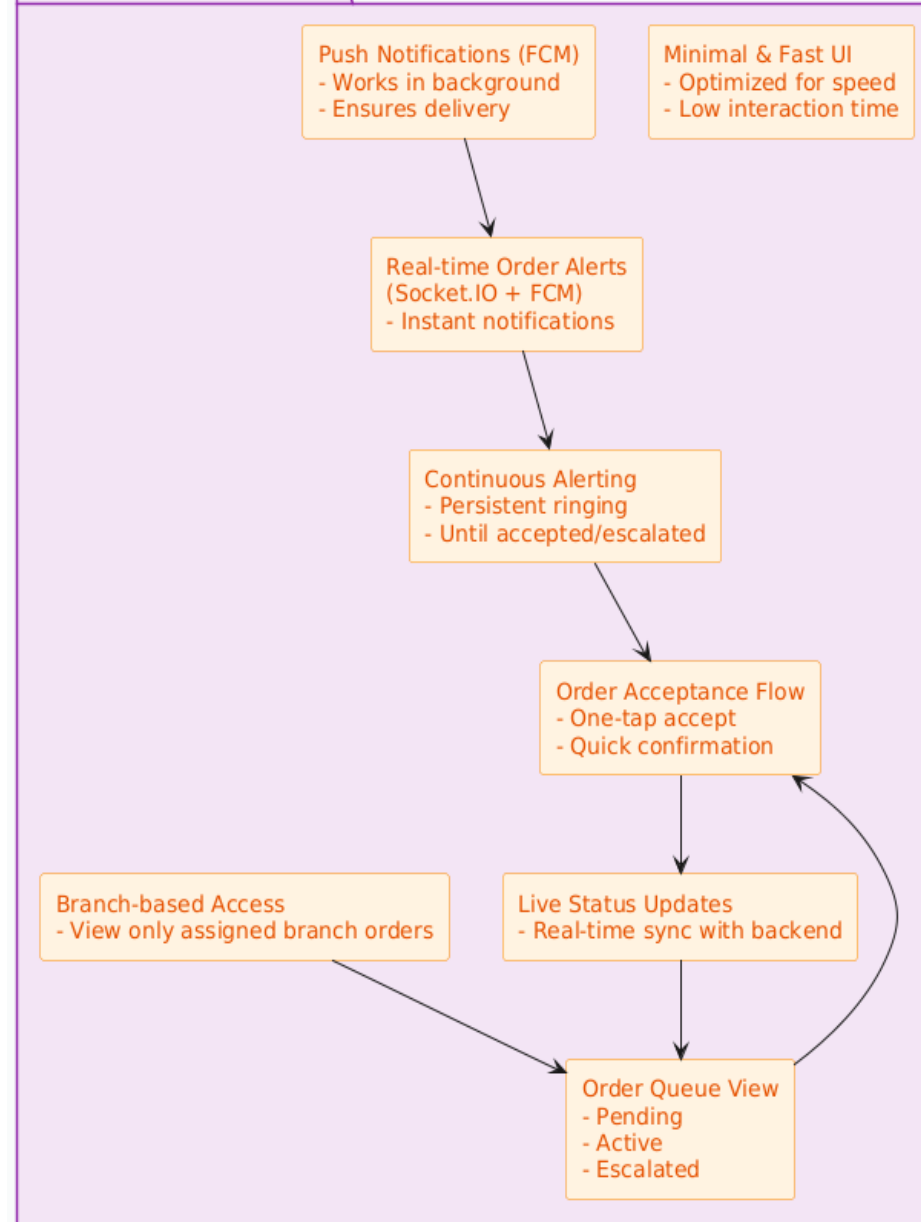


Mobile App Features

- Real-time Order Alerts
Instant notifications for new orders via Socket.IO + FCM
- Order Acceptance Flow
Quickly accept or confirm orders with one tap
- Continuous Alerting System
Persistent alerts until order is accepted or escalated
- Order Queue View
List of pending, active, and escalated orders
- Branch-based Access
Users only see orders assigned to their branch
- Push Notifications (FCM)
Ensures delivery even when app is closed or in background
- Live Status Updates
Real-time sync with backend after every action
- Minimal & Fast UI
Optimized for quick staff interaction during peak hours

Mobile App Features

Mobile App (React Native)



Advanced Server Log Monitoring

- Structured Logging System
Centralized logs with consistent format across all services
- Log Levels Classification
 - INFO: Normal system events (order received, notification sent)
 - SUCCESS: Successful operations (order accepted, FCM delivered)
 - WARN: Unexpected but non-critical issues (retry triggers, delays)
 - ERROR: Failures (failed notifications, DB errors, API failures)



Retry & Escalation Strategy

Step	Action
1	Send alert to kitchen devices
2	Wait 10-30 seconds
3	Retry (max 2-3 times)
4	Escalate to manager
5	Optional: Auto-assign



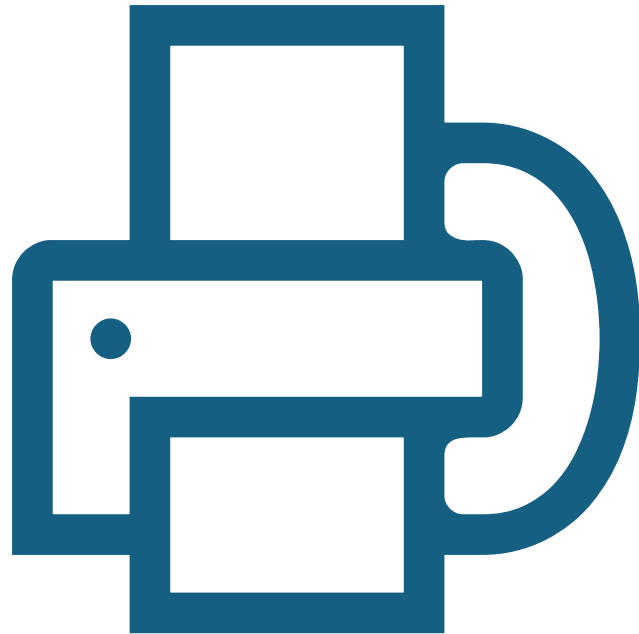
Integration Support from Tiffin

- ChilliPOS Integration
 - API Access & Documentation provided by Tiffin
- Food Delivery Platforms Integration
(*FoodPanda, Foodi, Pathao Food*)
 - Webhook Documentation & Setup Support
 - Assistance for configuration & testing



AI-DLC (AI-Driven Development Lifecycle)

- Context / Spec Driven Development
System behavior defined through structured context and specifications
- Minimal Manual Coding Approach
Aim to reduce handwritten code using AI-assisted generation
- AI-Assisted Implementation
Generate APIs, services, and components from defined specs
- Documentation-First Approach
Specs act as both documentation and development source



Support Required from Brain Station 23

- Hosting for Testing Environment
Provide VPS / server setup for staging (if not covered by client)
- Android Thermal Printer
Device support for receipt printing integration and testing
- Android Device + Wireless Thermal Printer
Required for end-to-end testing of mobile app and printing workflow